

RESEARCH PLAN + CURRENT EVIDENCE

The Verifier Bottleneck

When does self-improvement sharpen, expand, or collapse mathematical ability?

RESEARCH QUESTION

How do verifier quality and exploration affect self-improvement?

Dial 1: selection quality

How strongly does acceptance indicate correctness?

Dial 2: reachable candidates

How broadly can the policy search and recombine latent skills?

Sharpen

Increase first-try accuracy without reaching new problems.

Expand

Increase the set of problems reachable with many attempts.

Collapse

Train on misleading feedback and lose true accuracy.

Three hypotheses separate selection, exploration, and composition

H1 / SHARPENING

Without exploration, self-training only reallocates probability mass.

Prediction: pass@1 rises while large-k pass@k remains unchanged.

H2 / SELECTION

The verifier sets a collapse threshold.

Prediction: gains require $\alpha > \beta$, and grow with checker informativeness.

H3 / COMPOSITION

Exploration can create a capability absent from base samples.

Prediction: RL solves unseen compositions of separately learned skills.

Goals: derive the two-dial law (G1), map its phase diagram (G2), and catch skill composition in synthetic and real mathematics (G3).

A controlled affine-composition sandbox with an exact verifier

Task and training setup

$$g_i(x) = a_i x + b_i, \quad a_i, b_i \in \mathbb{F}_p$$

$$(g_n \circ \dots \circ g_1)(x) = Ax + B, \quad A, B \in \mathbb{F}_p$$

- Qwen3-0.6B-Base and Qwen3-1.7B-Base
- Standard on-policy GRPO with binary reward: $r = 1$ with probability α if correct and β if incorrect
- Depth: $n \sim \text{Unif}\{2, 3, 4, 5\}$
- Train primes: $p \in \{5, 7, 13, 19, 23, 29\}$
- Held-out primes: $p \in \{11, 17\}$

Scope: this tests H1, H2, G1, and G2 only. It does not yet test separately learned skills or H3.

Prompt and exact target

Compose linear functions modulo a prime p .
Start from $(A, B) = (1, 0)$ and update
 $A \leftarrow a*A, \quad B \leftarrow a*B + b \pmod{p}$.

Problem:

$p = 7$

$g_1(x) = 6*x + 4$

$g_2(x) = 6*x + 2$

Solution:

The full prompt contains two solved examples before the target problem.

Correct output

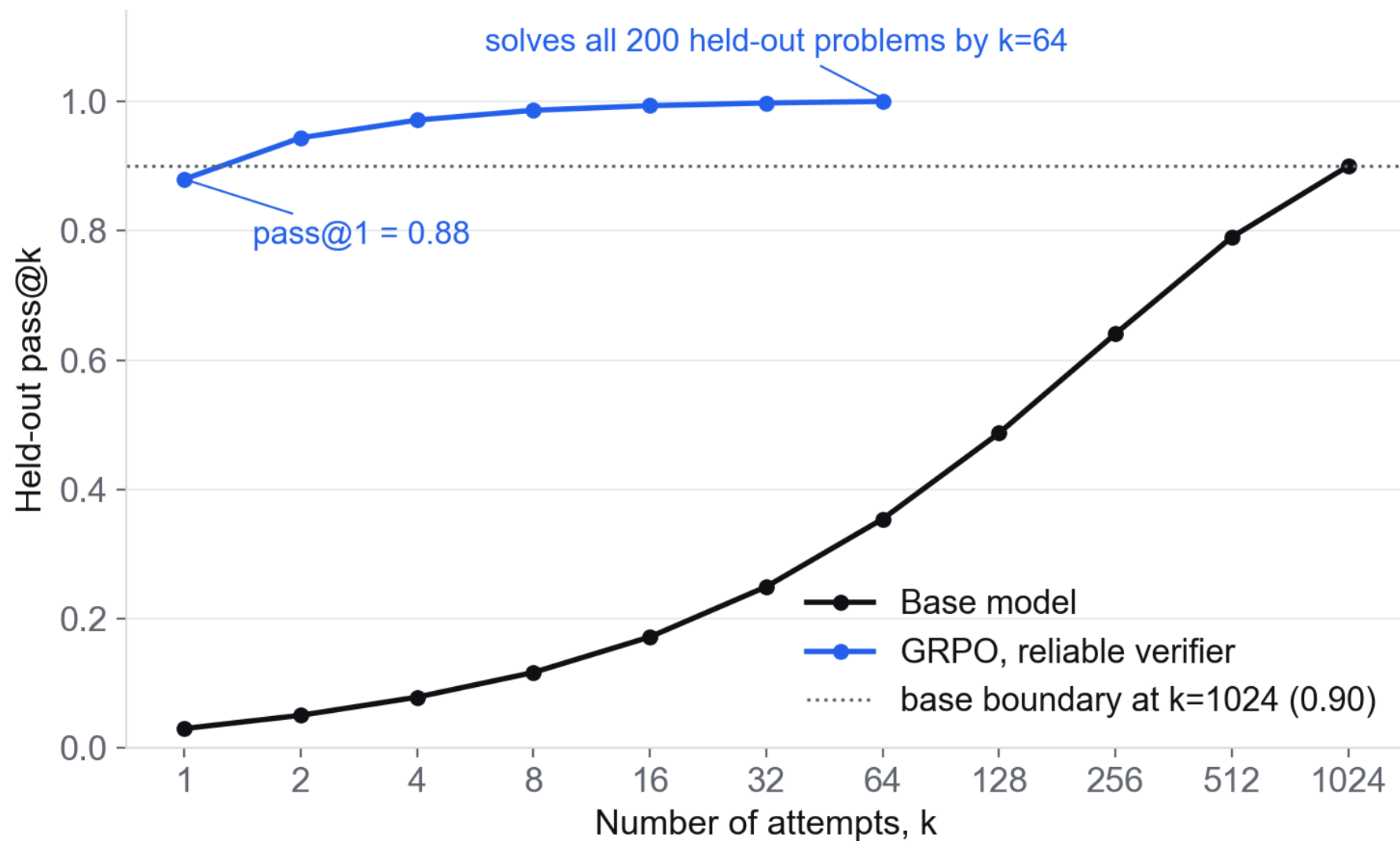
Start: $A=1, B=0$

g_1 : $A=6*1=6, B=6*0+4=4$

g_2 : $A=6*6=36=1, B=6*4+2=26=5$

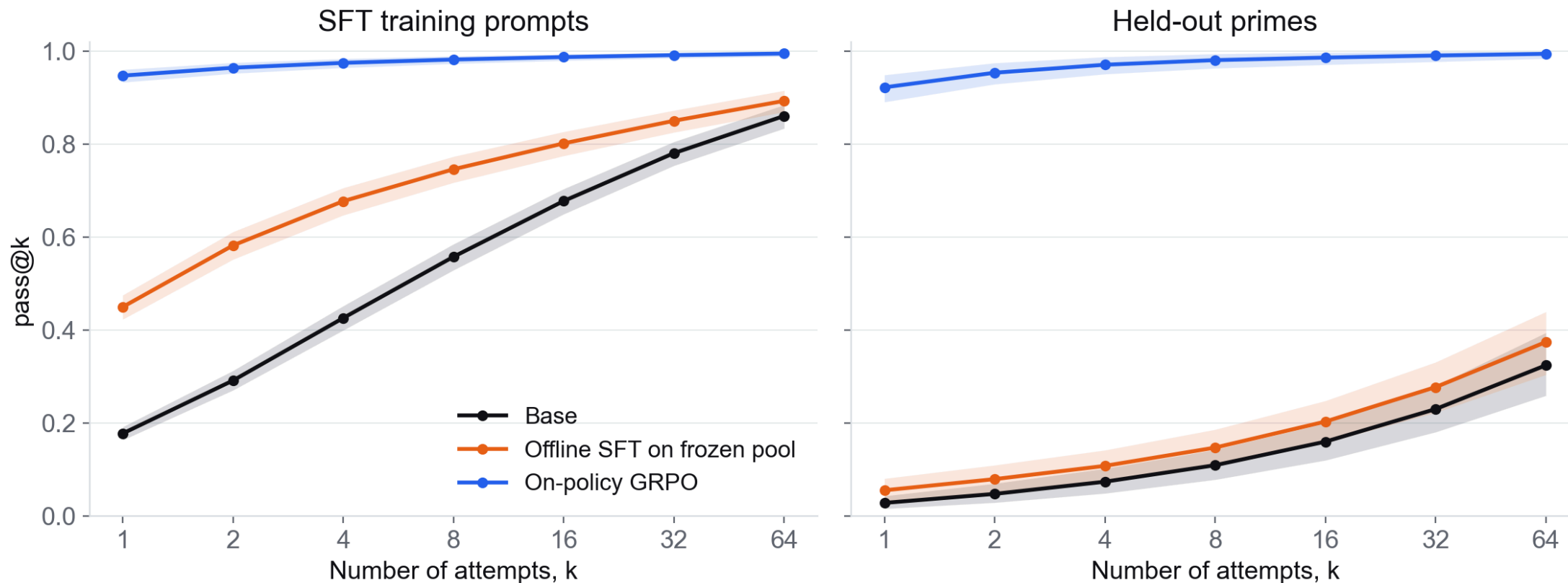
Answer: $A=1, B=5$

REFERENCE POINT / RELIABLE VERIFIER



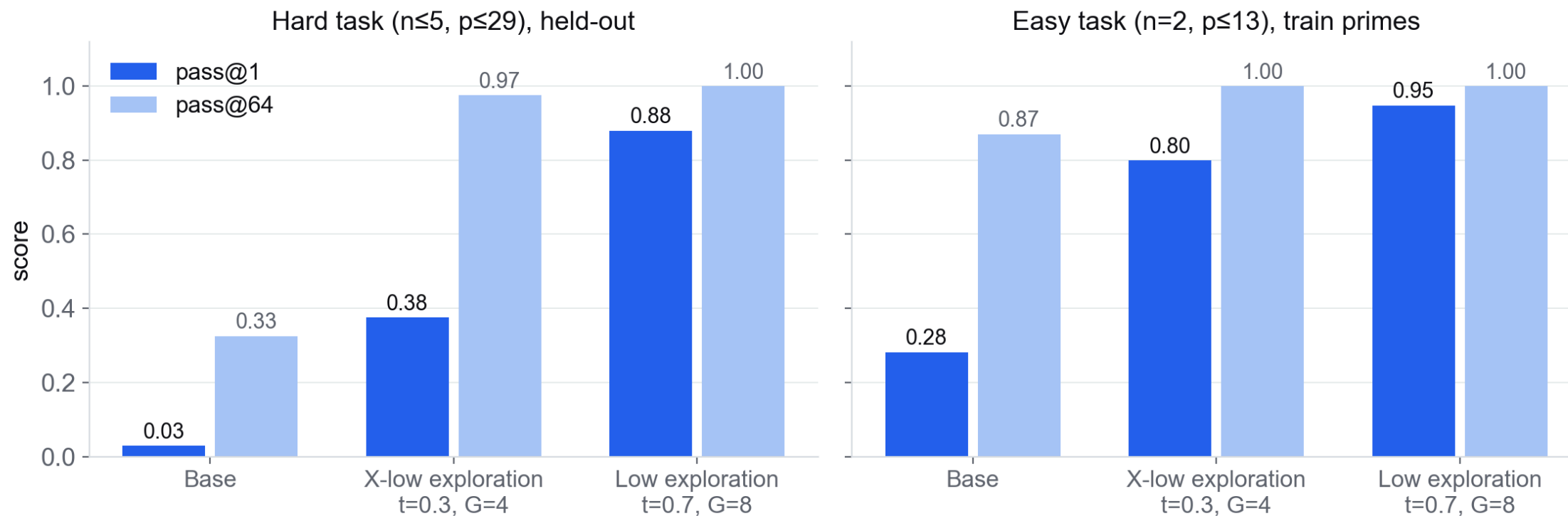
Offline reweighting sharpens; on-policy sampling expands

Design: same base pool and exact verifier. SFT trains on frozen samples; GRPO generates new candidates.



Result: SFT improves pass@1 but not pass@64; GRPO improves both. Training SFT 5× longer only memorizes the pool (pass@64 = 0.99 on its prompts) — on fresh problems pass@1 stays flat and pass@64 falls back to base.

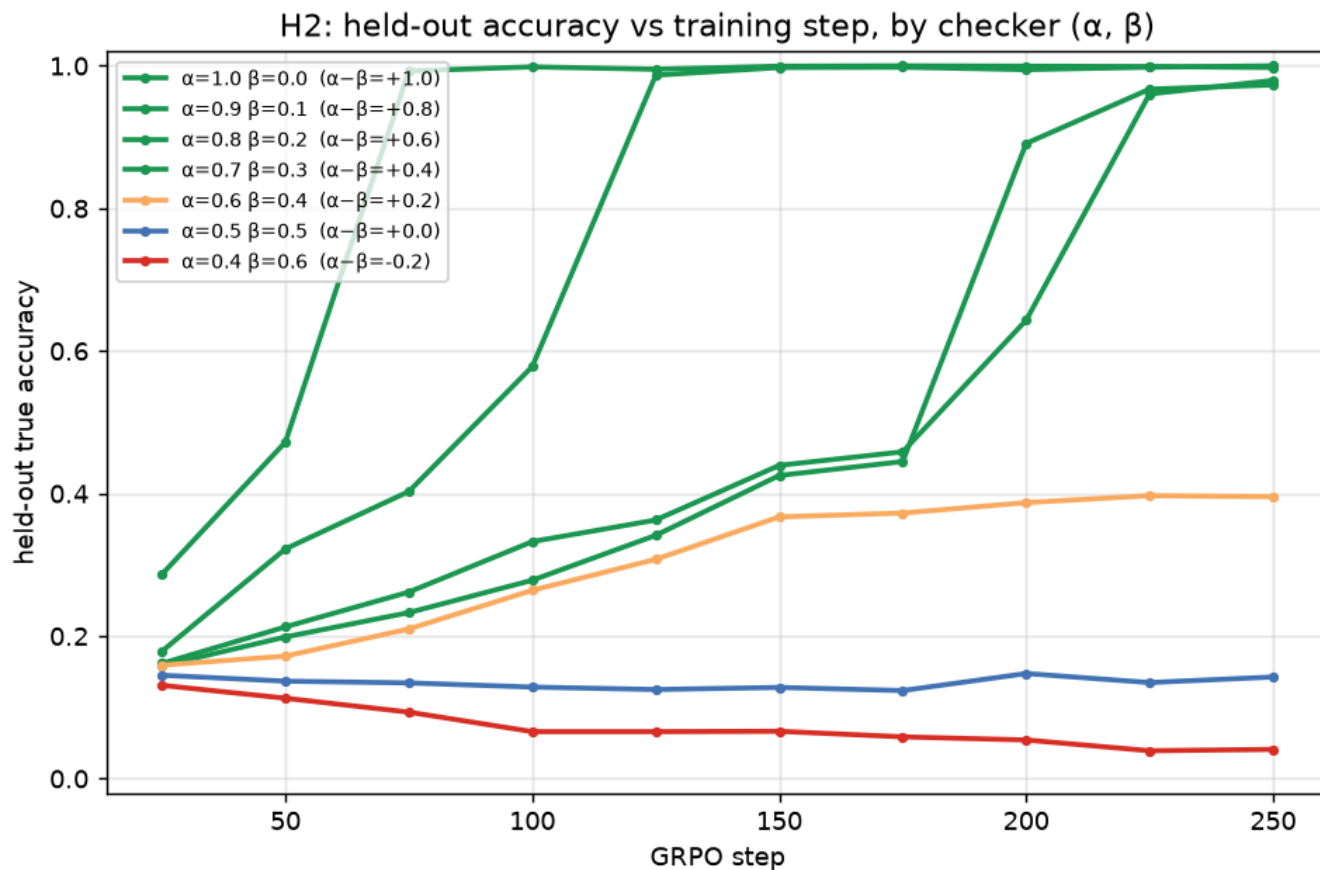
Even X-low exploration expands reach in the affine sandbox



Sandbox result: pass@1 and pass@64 rise even at $t = 0.3$, $G = 4$.

Open question: whether this persists on other tasks.

Checker sign sets direction; quality sets convergence speed



$\alpha > \beta$: held-out true accuracy rises, but weakly informative checker settings need substantially more steps to approach perfect quality.

$\alpha = \beta$: accuracy stalls through 250 steps; a longer run suggests delayed degradation.

$\alpha < \beta$: accuracy falls.

Pilot only: Qwen3-1.7B-Base, depth 2, one seed, and the single slice $\alpha + \beta = 1$.

Run the full $\alpha \times \beta$ grid long enough to reach convergence

5 × 5 checker sweep

	$\beta = 0$	0.2	0.4	0.6	0.8
$\alpha = 1.0$	•	•	•	•	•
0.8	•	•	•	•	•
0.6	•	•	•	•	•
0.4	•	•	•	•	•
0.2	•	•	•	•	•

Fixed model, task distribution, exploration, and generations per step. Three seeds per cell; at least 600 steps, extended until learning curves plateau.

Questions this grid answers

Boundary: after convergence, is $\alpha = \beta$ the true divide between learning and collapse?

Speed: within $\alpha > \beta$, how quickly does each checker reach 90–95% quality?

Error mix: at equal $\alpha - \beta$, are false positives more damaging than false negatives?

Ceiling: do noisy positive-margin checkers eventually match the perfect checker, or plateau lower?

Track: held-out true pass@1 over training, time to target quality, and final forgetting of base-solved problems.

* A 150-step sweep was run, but this horizon is too short: many $\alpha > \beta$ cells were still improving. We do not treat its endpoint comparisons as valid results.

Does atomic RL transfer to unseen compositions?

Let $P(x) = \sum_{r=0}^d a_r x^r \in \mathbb{F}_p[x]$ be a polynomial over the finite field $\mathbb{F}_p$.

SHIFT	DERIVATIVE	SQUARE + TRUNCATE	...
$A_1[P](x) = P(x + 2)$	$A_2[P](x) = P'(x)$	$A_3[P](x) = P(x)^2 \bmod x^{d+1}$	

Experiment design

- 01 Choose one pair $\{A_i, A_j\}$ and start from the base checkpoint.
- 02 Train one model with RL on fresh $A_i[P]$ and $A_j[P]$ tasks simultaneously, sampled 50/50.
- 03 Never train or reward $A_i(A_j[P])$ or $A_j(A_i[P])$.

Repeat for all $\binom{n}{2}$ pairs and at least three seeds.

Measure before and after RL

TASK	ROLE	METRICS
$A_i[P], A_j[P]$	trained	pass@1, pass@64
$A_i(A_j[P])$	unseen	pass@1, pass@64, pass@1024
$A_j(A_i[P])$	unseen	pass@1, pass@64, pass@1024

H3: learning the two atoms increases reach on either unseen composition.